

Honours Probability - MATH 458

Professor Chen - lichen@math.mcgill.ca

Office Hours: TBA - Burn 1230

Contents

1	The Basics of Probability	1
1.1	Overview of a Probability Space	1
1.2	Conditional Probability	9
1.3	Independence of Events	12
2	Discrete and Continuous Random Variables	14
2.1	Distributions	14
2.2	Random Variables	18
2.2.1	Discrete Random Variables	18
2.2.2	Expectation of a Discrete Random Variable	23
3	Variance, Moments, and Generating Functions	28
4	Multiple Random Variables: Correlation, Independence, Conditioning	28
5	The Law of Large Numbers	28

1 The Basics of Probability

1.1 Overview of a Probability Space

A probability space $(\Omega, \mathcal{S}, \mathbb{P})$ is a model of random trials/experiments. This means

- all possible outcomes are known: e.g. flipping a coin.
- the outcome of any specific performance is unknown
- repeatable under identical conditions

Definition 1.1. $\Omega = \{\text{all possible outcomes}\}$ is the sample space. $\omega \in \Omega$ is called a sample point.

Definition 1.2. $\mathcal{S} = \{\text{all the events we can study}\}$ is a σ -field. $\mathcal{S}$ is a collection of subsets of Ω . $A \in \mathcal{S}$ is an event. $\emptyset, \Omega$ are the trivial events.

Definition 1.3. $\mathbb{P} : A \in \mathcal{S} \mapsto [0, 1]$ with the requirement that $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$ is the probability.

Example 1.4. Flip a coin once: $\Omega = \{H, T\}; \mathcal{S} = \{\{H\}, \{T\}, \{H, T\}, \emptyset\} = 2^\Omega := \{\text{all subsets of } \Omega\}$.

$$\mathbb{P}(\{H\}) = \frac{1}{2} = \mathbb{P}(\{T\})$$

Example 1.5. Roll a fair die once: $\Omega = \{1, 2, 3, 4, 5, 6\}; \mathcal{S} = 2^\Omega$.

$$\mathbb{P}(\{1\}) = \frac{1}{6}, \mathbb{P}(\{2, 3\}) = \frac{1}{3}$$

Example 1.6. Choose any number at random from $[0, 1]$. $\Omega = [0, 1]$.

$$\mathbb{P}\left(\left\{\frac{1}{2}\right\}\right) = 0, \mathbb{P}\left(\left[0, \frac{1}{2}\right]\right) = \frac{1}{2}$$

Example 1.7. Consider $\Omega = \mathbb{N}, \mathcal{S} = 2^\mathbb{N}$. $\mathbb{P}(n) = 2^{-n}$.

Definition 1.8. $\mathcal{S}$ is a collection of subsets of Ω . $\mathcal{S}$ is called a σ -field if

- $\Omega \in \mathcal{S}$
- if $A \in \mathcal{S}$ then $A^c \in \mathcal{S}$
- if $A_i \in \mathcal{S} \forall i \in \mathbb{N}$ then $\cup_i A_i \in \mathcal{S}$

Some results of this are

- $\emptyset \in \mathcal{S}$
- if $A_i \in \mathcal{S}, i \in \{1, \dots, N\}$ then $\cup_{i=1}^N A_i \in \mathcal{S}, \cap_{i=1}^N A_i \in \mathcal{S}$
- if $A_i \in \mathcal{S} \forall i \in \mathbb{N}$ then $\cap_i A_i \in \mathcal{S}$
- if $A, B \in \mathcal{S}$ then $B \setminus A \in \mathcal{S}$

Definition 1.9. Let C be a collection of subsets of Ω . The σ -field generated by C , denoted $\sigma(C)$ is the smallest σ -field that contains C , i.e., if $\mathcal{S}$ is a σ -field and $C \subset \mathcal{S}$ then $\sigma(C) \subset \mathcal{S}$.

Remark 1.10. For $A \in \mathcal{S}$ the smallest σ -field containing A , i.e., the σ -field generated by A is $\mathcal{S} = \{\emptyset, \Omega, A, A^c\}$.

Remark 1.11. For $A, B \in \mathcal{S}$ the smallest σ -field containing A, B , i.e., the σ -field generated by A, B is $\mathcal{S} = \{\emptyset, \Omega, A, A^c, B, B^c, A \cup B, A \cap B, A \cup B^c, A^c \cup B, (A \cup B)^c, (A \cup B)^c \cup (A \cap B)\}$.

Proposition 1.12.

$$\sigma(C) = \cap \{ \mathcal{S} : \mathcal{S} \text{ is a } \sigma\text{-field and } C \subset \mathcal{S} \}$$

Note that any intersection of σ -fields is a σ -field.

Proof. By definition of $\sigma(C)$, $\sigma(C)$ is an element of the $\{ \mathcal{S} : \mathcal{S} \text{ is a } \sigma\text{-field and } C \subset \mathcal{S} \}$, so we have the inclusion $\supset$.

We also have that the right hand side is a σ -field that contains C so we also have the opposite inclusion, since $\sigma(C)$ is the smallest of these. $\square$

Remark 1.13. If $\mathcal{S}_1, \mathcal{S}_2$ are σ -fields then $\mathcal{S}_1 \cup \mathcal{S}_2$ is not necessarily a σ -field.

Definition 1.14. The Borel σ -field on $\mathbb{R}$, denoted by $\mathcal{B}(\mathbb{R})$, is the σ -field generated by all the open intervals on $\mathbb{R}$, i.e., $\mathcal{B} = \sigma(\{(a, b) : a < b; a, b \in \mathbb{R}\})$.

Remark 1.15. • $\forall a \in \mathbb{R} : (a, \infty), (-\infty, a) \in \mathcal{B}(\mathbb{R})$

- $\forall a \in \mathbb{R} : \{a\} \in \mathcal{B}(\mathbb{R})$
- Any interval is in $\mathcal{B}(\mathbb{R})$

Remark 1.16. There exist subsets of $\mathbb{R}$ that are not Borel.

Definition 1.17. Let Ω be a sample space and $\mathcal{S}$ a σ -field on Ω . A set function $\mathbb{P}$ defined on $\mathcal{S}$ is a probability (probability measure) if

- $\forall A \in \mathcal{S}, \mathbb{P}(A) \geq 0$
- $\mathbb{P}(\Omega) = 1$
- It has countable additivity, i.e., if $A_i \in \mathcal{S}; i \in \mathbb{N}$; and $\forall i \neq j, A_i \cap A_j = \emptyset$ then $\mathbb{P}(\cup_i A_i) = \sum_i \mathbb{P}(A_i)$

Remark 1.18. Some properties of $\mathbb{P}$ are

- $\mathbb{P}(\emptyset) = 0. \mathbb{P}(\emptyset) = \sum_{i=1}^{\infty} \mathbb{P}(\emptyset) \implies \mathbb{P}(\emptyset) = 0.$
- Finite Additivity
- $\forall A \in \mathcal{S}, \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- if $A, B \in \mathcal{S}$ and $A \subset B$ then $\mathbb{P}(B - A) = \mathbb{P}(B) - \mathbb{P}(A)$. Note that $\mathbb{P}(A) < \mathbb{P}(B)$.
- if $A, B \in \mathcal{S}$ then $\mathbb{P}(B \cup A) = \mathbb{P}(B) + \mathbb{P}(A) - \mathbb{P}(A \cap B)$. This is the addition rule. Note that this implies finite subadditivity, i.e., $\mathbb{P}(\cup_i A_i) \leq \sum_i \mathbb{P}(A_i)$

- (Inclusion-Exclusion Formula) if $A_i \in \mathcal{S}, i \in \{1, \dots, N\}$ then

$$\begin{aligned} & \mathbb{P}\left(\bigcup_{i=1}^N A_i\right) \\ &= \sum_{i=1}^N \mathbb{P}(A_i) - \sum_{1 \leq i_1 < i_2 \leq N} \mathbb{P}(A_{i_1} \cap A_{i_2}) + \dots + (-1)^{N+1} \mathbb{P}\left(\bigcap_{i=1}^N A_i\right) \\ &= \sum_{k=1}^N (-1)^{i+1} \sum_{1 \leq i_1 < \dots < i_k \leq N} \mathbb{P}\left(\bigcap_{n=i_1}^{i_k} A_n\right) \end{aligned}$$

- (Bonferroni's Inequality) if $A_i \in \mathcal{S}, i \in \{1, \dots, N\}$ then

$$\sum_{i=1}^N \mathbb{P}(A_i) - \sum_{i_1 < i_2} \mathbb{P}(A_{i_1} \cap A_{i_2}) \leq \mathbb{P}\left(\bigcup_{i=1}^N A_i\right) \leq \sum_{i=1}^N \mathbb{P}(A_i)$$

Proof. (Inclusion-Exclusion Formula) For the base case of 1 set the statement is just $\mathbb{P}(A) = \mathbb{P}(A)$. Suppose it is true for $1, \dots, n$ sets. Then by the addition rule

$$\mathbb{P}\left(\bigcup_{i=1}^{n+1} A_i\right) = \mathbb{P}(A_{n+1}) + \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) - \mathbb{P}\left(\bigcup_{i=1}^n (A_i \cap A_{n+1})\right)$$

Applying the on the second and third terms assumption the the formula holds for n sets and then collecting terms we finally have

$$= \sum_{k=1}^{n+1} (-1)^{i+1} \sum_{1 \leq i_1 < \dots < i_k \leq n+1} \mathbb{P}\left(\bigcap_{j=i_1}^{i_k} A_j\right)$$

□

Proof. (Bonferroni's Inequality)

The right inequality is obvious. The statement is obvious for the base case of 1 set. Now assume it holds for N

$$\begin{aligned} & \mathbb{P}(\cup_{i=1}^{N+1} A_i) = \mathbb{P}((\cup_{i=1}^N A_i) \cup A_{N+1}) = \mathbb{P}(\cup_{i=1}^N A_i) + \mathbb{P}(A_{N+1}) - \mathbb{P}((\cup_{i=1}^N A_i) \cap A_{N+1}) \\ & \leq \sum_{i=1}^{N+1} \mathbb{P}(A_i) - \sum_{i_1 < i_2} \mathbb{P}(A_{i_1} \cap A_{i_2}) - \sum_{i=1}^N \mathbb{P}(A_i \cap A_{N+1}) = \sum_{i=1}^{N+1} \mathbb{P}(A_i) - \sum_{i_1 < i_2} \mathbb{P}(A_{i_1} \cap A_{i_2}) \end{aligned}$$

□

Remark 1.19. Ω is a sample space, $\omega \in \Omega$. $\{\omega\}$ is a singleton. If $\{\omega\} \in \mathcal{S}$ then it is an elementary event.

- Ω is discrete and finite. $\Omega = \{\omega_1, \dots, \omega_n\}$ and $\mathcal{S} = 2^\Omega$.

It is sufficient to assign $\mathbb{P}$ on every elementary event, i.e., $\mathbb{P}(\{\omega_j\}), j = 1, \dots, n$. Take the probability of any set A to be the sum of its elementary events.

One such $\mathbb{P}$ is the uniform probability with $\mathbb{P}(\{\omega_j\}) = \frac{1}{n}$ for every j .

- $\Omega = \{\omega_1, \omega_2, \dots\}$ (countably infinite), $\mathcal{S} = 2^\Omega$

One cannot assign a uniform probability on each elementary event. However one can assign $\mathbb{P}(\{\omega_j\}) = 2^{-j}$ or

$$\mathbb{P}(\{\omega_j\}) = \begin{cases} 1 & \text{if } j = 1 \\ 0 & \text{otherwise} \end{cases} \quad (1.1)$$

- $\Omega = [0, 1], \mathcal{S} = \mathcal{B}([0, 1])$

Exercise: if Ω is uncountable and $\forall \omega \in \Omega, \{\omega\} \in \mathcal{S}$ then for any probability $\mathbb{P}$ on $(\Omega, \mathcal{S})$ there are at most countable ω such that $\mathbb{P}(\{\omega\}) > 0$.

Proof. Consider the sets $A_n = \{\omega \in \mathcal{S} : \mathbb{P}(\{\omega\}) > \frac{1}{n}\}$. Each set has less than n elements, otherwise $\mathbb{P}(A_n) \geq n \cdot \frac{1}{n} = 1$. $\bigcup_{n=1}^{\infty} A_n$ is the set of all events with nonzero probability. It is a countable union of finite sets and is thus countable. Therefore, there are at most countable events with nonzero probability. $\square$

Definition 1.20. Assume that $A_i \in \mathcal{S}, i \in \mathbb{N}$. $\{A_i : i \in \mathbb{N}\}$ is a non-decreasing sequence if $A_i \subset A_{i+1}, \forall i \in \mathbb{N}$. It is non-increasing if $A_i \supset A_{i+1}, \forall i \in \mathbb{N}$. If it is non-increasing or non-decreasing then it is monotone.

If $\{A_i : i \in \mathbb{N}\}$ is monotone then we have existence of $\lim_{i \rightarrow \infty} A_i = \bigcup_{i=1}^{\infty} A_i$ or $\bigcap_{i=1}^{\infty} A_i$ for non-decreasing and non-increasing respectively.

Example 1.21. $a \in \mathbb{R}, A_i = (a - \frac{1}{i}, a + \frac{1}{i})$ then $\{A_i : i \in \mathbb{N}\}$ is non-increasing and $\lim_{i \rightarrow \infty} A_i = \{a\}$.

Example 1.22. $A_i = (-i, i)$ then $\{A_i : i \in \mathbb{N}\}$ is non-decreasing and $\lim_{i \rightarrow \infty} A_i = \mathbb{R}$.

Remark 1.23. Some more properties of $\mathbb{P}$ are

- if $\{A_i : i \in \mathbb{N}\} \subset \mathcal{S}$ is non-decreasing then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \lim_{i \rightarrow \infty} \mathbb{P}(A_i) \iff \mathbb{P}(\lim_i A_i) = \lim_i \mathbb{P}(A_i)$. (continuity from below)
- $\{A_i : i \in \mathbb{N}\} \subset \mathcal{S}$ a non-increasing sequence then $\mathbb{P}(\bigcap_{i=1}^{\infty} A_i) = \lim_i \mathbb{P}(A_i) \iff \mathbb{P}(\lim_i A_i) = \lim_i \mathbb{P}(A_i)$. Note: Proof is from the fact that A_i^c is a non-decreasing sequence.

- $\{A_i : i \in \mathbb{N}\} \subset \mathcal{S}$ then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$. (countable subadditivity)

Proof. (of continuity from below)

Define $B_1 = A_1, B_i = \frac{A_i}{A_{i-1}}, i \geq 2$. Because $A_i \uparrow$, $\{B_i\}$ are disjoint

$$\forall n \in \mathbb{N}, \bigcup_{i=1}^n B_i = \bigcup_{i=1}^n A_i = A_n$$

We know

$$\begin{aligned} \mathbb{P}\left(\bigcup_{i=1}^{\infty} B_i\right) &= \sum_{i=1}^{\infty} \mathbb{P}(B_i) = \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbb{P}(B_i) \\ &= \lim_{n \rightarrow \infty} \mathbb{P}\left(\bigcup_{i=1}^n B_i\right) = \lim_{n \rightarrow \infty} \mathbb{P}(A_n) \\ &\implies \mathbb{P}\left(\lim_i A_i\right) = \lim_i \mathbb{P}(A_i) \end{aligned}$$

□

Proof. (countable subadditivity)

Define $B_n = \bigcup_{i=1}^n A_i, \forall i \in \mathbb{N}$.

$$\begin{aligned} B_n \uparrow &\implies \lim_n B_n = \bigcup_{n=1}^{\infty} B_n = \bigcup_{i=1}^{\infty} A_i \\ &\implies \mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \lim_n \mathbb{P}(B_n) \\ \mathbb{P}(B_n) &= \mathbb{P}\left(\bigcup_{i=1}^n A_i\right) \leq \sum_{i=1}^n \mathbb{P}(A_i) \implies \lim_n \mathbb{P}(B_n) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i) \end{aligned}$$

□

Remark 1.24. $\Omega = \{\omega_1, \dots, \omega_n\}$ is discrete and finite, $\mathcal{S} = 2^{\Omega}$. It is sufficient to assign $\mathbb{P}$ on $\{\omega_j\}, j = 1, \dots, n$. One example is the uniform probability: $\mathbb{P}(\{\omega_j\}) = \frac{1}{n}$.

Example 1.25. (Matching Problem)

Suppose there are m men at a party throwing their hats into a basket. The hats are mixed up and then each man randomly selects a hat. What is the probability that none of the m men selects their own hat?

$\Omega = \{(a_1, \dots, a_i, \dots, a_m) : a_i \in \{1, \dots, m\}, i = 1, \dots, m\}$ e.g. $m = 5, (3, 1, 4, 5, 2)$ (men and hats each numbered from 1 to m). This is a permutation, $|\Omega| = m!$. $\mathbb{P}$ is uniform. Define $E_i = \{(a_1, \dots, a_m) : a_i = i\}$.

$|E_i| = (m-1)!$ (permutations of $m-1$ elements) $\mathbb{P}(E_i) = \frac{(m-1)!}{m!}$

$$E = \left(\bigcup_{i=1}^m E_i \right)^c$$

Inclusion exclusion formula:

$$\mathbb{P} \left(\bigcup_{i=1}^m E_i \right) = \sum_{i=1}^m \mathbb{P}(E_i) - \sum_{i_1 < i_2} \mathbb{P}(E_{i_1} \cap E_{i_2}) + \dots + (-1)^{m+1} \mathbb{P} \left(\bigcap_{i=1}^m E_i \right)$$

For $i_1 < \dots < i_n, |E_{i_1} \cap \dots \cap E_{i_n}| = (m-n)! \implies \frac{(m-n)!}{m!}$.

$$|\{(i_1, \dots, i_m) : i_1 < \dots < i_m\}| = \binom{m}{n} = \frac{m!}{n!(m-n)!}$$

$$\begin{aligned} \mathbb{P} \left(\bigcup_{i=1}^m E_i \right) &= m \frac{(m-1)!}{m!} - \binom{m}{2} \frac{(m-2)!}{m!} + \dots + (-1)^{m+1} \frac{1}{m!} \\ &= \sum_{i=1}^m (-1)^{i+1} \binom{m}{i} \frac{(m-i)!}{m!} = \sum_{i=1}^m (-1)^{i+1} \frac{1}{i!} \end{aligned}$$

$$\mathbb{P}(E) = 1 - \sum_{i=1}^m (-1)^{i+1} \frac{1}{i!}$$

What if $m \rightarrow \infty$?

$$\begin{aligned} 1 - \lim_{m \rightarrow \infty} \left(\sum_{i=1}^m (-1)^{i+1} \frac{1}{i!} \right) &= 1 + \lim_{m \rightarrow \infty} \sum_{i=1}^m (-1)^i \frac{1}{i!} = \lim_{m \rightarrow \infty} \sum_{i=0}^m (-1)^i \frac{1}{i!} \\ &= \sum_{i=0}^{\infty} \frac{(-1)^i}{i!} = e^{-1} \approx 0.37 \end{aligned}$$

Note: combinatorics is Section 1.4 of R-S.

Remark 1.26. For $\Omega = [0, 1], \mathbb{R}, \mathbb{R}^n, [a, b] \times [c, d], \mathcal{S} = \mathcal{B}(\Omega)$ we do have a uniform probability.

$\Omega = [0, 1]$, uniform $\mathbb{P}$ is such that for every interval $I = [a, b] \in [0, 1], \mathbb{P}(I) = |b - a|$. ($\mathbb{P}([0, \frac{1}{2}]) = \frac{1}{2}$).

$\Omega = D \subset \mathbb{R}^n$ bounded, uniform $\mathbb{P}$ is such that $\forall A \in \mathcal{B}(D), \mathbb{P} = \frac{\text{vol} A}{\text{vol} D}$.

Remark 1.27. A countable union of null sets ($A \in \mathcal{S}, \mathbb{P}(A) = 0$ then A is a null set) is again a null set.

Example 1.28. Flip a fair coin infinitely many times. What is the probability that successive heads never occur?

Proof. Consider the first n flips. Define A_n to be the event that no successive heads occur. A_n^c is the event that successive heads occur. Define a_n to be the number of ways of flipping a coin n times but successive heads do not occur.

$$a_n = |A_n|; \quad \mathbb{P}(A_n) = \frac{a_n}{2^n}$$

$$a_1 = 2, a_2 = 3, \dots$$

Now suppose that the first flip is a tails. Then for A_n to occur, there must be no successive heads in the other $n - 1$ flips. Otherwise heads is flipped first. Then tails must be flipped next and the $n - 2$ other flips must not have successive heads. So we have the Fibonacci sequence

$$a_n = a_{n-1} + a_{n-2}$$

note that (a_n) is increasing. Then we have

$$a_n = a_{n-1} + a_{n-2} \leq 2a_{n-1} \implies a_n \leq 3a_{n-2}$$

If $n = 2k$ then we have

$$a_n \leq 3^{k-1}a_2 = 3^k = (\sqrt{3})^n$$

If $n = 2k + 1$ then

$$\begin{aligned} a_n &= 3^k a_1 = 2 \cdot 3^k = \frac{2}{\sqrt{3}} \cdot (\sqrt{3})^n \\ \implies \frac{a_n}{2^n} &\leq \frac{\frac{2}{\sqrt{3}}(\sqrt{3})^k}{2^n} \xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

□

Example 1.29. Choose two points randomly from $[0, 1]$. The two points divide $[0, 1]$ into three line segments. What is the probability that the three segments form a triangle? (event A)

Proof.

$$\Omega = \{(x, y) : x, y \in [0, 1]\} = [0, 1] \times [0, 1]; \quad \mathcal{S} = \mathcal{B}(\Omega)$$

If $x < y$ then we must have $x < (y - x) + (1 - y); y - x < x + (1 - y); 1 - y < x + (y - x)$. This gives us the requirement that $x < \frac{1}{2}; y < x + \frac{1}{2}; y > \frac{1}{2}$. Then we have that (x, y) is in the upper left quadrant of $[0, 1]^2$ and also below the line $y = x + \frac{1}{2}$.

Now if $x > y$ we have the same argument in reverse. Then we have two disjoint right triangles with sidelengths $\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}$ so we have $\mathbb{P}(A) = \frac{2 \cdot \frac{1}{2} \cdot (\frac{1}{2})^2}{1} = \frac{1}{4}$ □

1.2 Conditional Probability

The study of $(\Omega, \mathcal{S}, \mathbb{P})$ where conditioning on H occurs for some $H \in \mathcal{S}$.

Example 1.30. Toss a fair coin twice.

$$\Omega = \{HH, HT, TH, TT\}$$

$\mathbb{P}$ is the equal probability function.

Now toss a fair coin twice and suppose that H occurs at least once.

$$\Omega = \{HH, HT, TH\}; \quad \mathbb{P}(HH) = \frac{1}{3}$$

Definition 1.31. Let $(\Omega, \mathcal{S}, \mathbb{P})$ be a probability space and let $H \in \mathcal{S}, \mathbb{P}(H) > 0$. Then, $\forall A \in \mathcal{S}$, define

$$\mathbb{P}(A|H) := \frac{\mathbb{P}(A \cap H)}{\mathbb{P}(H)}$$

to be the conditional probability of A conditioning on H . If $\mathbb{P}(H) = 0$ then $\mathbb{P}(\cdot|H)$ is undefined.

Theorem 1.32. Let $(\Omega, \mathcal{S}, \mathbb{P})$ and H be the same as above. Then the set function $\mathbb{P}(\cdot|H) : A \in \mathcal{S} \mapsto \mathbb{P}(A|H)$ is a probability on $(\Omega, \mathcal{S})$.

Proof. First, $\forall A \in \mathcal{S}, \mathbb{P}(A|H) \geq 0$ and $\mathbb{P}(\Omega|H) = \frac{\mathbb{P}(\Omega \cap H)}{\mathbb{P}(H)} = 1$. Now take $\{A_i : i \in \mathbb{N}\}$ to be a sequence of disjoint events

$$\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i|H\right) = \frac{\mathbb{P}\left(\left(\bigcup_{i=1}^{\infty} A_i\right) \cap H\right)}{\mathbb{P}(H)} = \frac{\mathbb{P}\left(\bigcup_{i=1}^{\infty} (A_i \cap H)\right)}{\mathbb{P}(H)} = \sum_{i=1}^{\infty} \frac{\mathbb{P}(A_i \cap H)}{\mathbb{P}(H)}$$

□

Corollary 1.33. Let $(\Omega, \mathcal{S}, \mathbb{P})$ and H be the same as above. Define

$$\mathcal{S}_H = \{A \cap H : A \in \mathcal{S}\}$$

Then $(H, \mathcal{S}_H, \mathbb{P}(\cdot|H))$ is a probability space.

Proof. Exercise. Note: show $\mathcal{S}_H$ is a σ -field, show $\mathbb{P}(\cdot|H)$ is a probability on $(H, \mathcal{S}_H)$. □

Remark 1.34.

$$\mathbb{P}(A|H) = \frac{\mathbb{P}(A \cap H)}{\mathbb{P}(H)} \iff \mathbb{P}(A \cap H) = \mathbb{P}(H)\mathbb{P}(A|H)$$

Theorem 1.35. (*The Multiplication Rule*)

If $A_i \in \mathcal{S}, i = 1, 2, \dots, n$ and $\mathbb{P}(\bigcap_{i=1}^n A_i) > 0$. Then

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \mathbb{P}(A_1)\mathbb{P}(A_2|A_1)\mathbb{P}(A_3|A_2 \cap A_1)\dots = \prod_{i=1}^n \mathbb{P}\left(A_i \mid \bigcap_{j=1}^{i-1} A_j\right)$$

Proof.

$$\mathbb{P}\left(\bigcap_{j=1}^{n-1} A_j\right) > 0 \implies \mathbb{P}\left(\bigcap_{j=1}^k A_j\right) > 0 \forall k = 1, 2, \dots, n-1$$

$$\prod_{i=1}^n \mathbb{P}\left(A_i \mid \bigcap_{j=1}^{i-1} A_j\right) = \mathbb{P}(A_1) \frac{\mathbb{P}(A_1 \cap A_2)}{\mathbb{P}(A_1)} \dots \frac{\mathbb{P}(\bigcap_{i=1}^n A_i)}{\mathbb{P}(\bigcap_{i=1}^{n-1} A_i)} = \mathbb{P}\left(\bigcap_{i=1}^n A_i\right)$$

□

Example 1.36. Recall the matching problem. Probability of no men receiving their hat is $\sum_{i=0}^m \frac{(-1)^i}{i!}$. What is the probability that exactly k of the m men receive their original hat?

Proof. Fix k spots out of M . Define E to be that event that each man of this set of k men gets his own hat. Define G to be the event that none of the rest of the $(m-k)$ gets his hat back. We then want to find

$$\mathbb{P}(E \cap G) = \mathbb{P}(E)\mathbb{P}(G|E) = \frac{(m-k)!}{m!} \cdot \sum_{i=0}^{m-k} \frac{(-1)^i}{i!}$$

$$\mathbb{P}(\text{exactly } K \text{ of } m \text{ men have matches}) = \binom{m}{k} \frac{(m-k)!}{m!} \cdot \sum_{i=0}^{m-k} \frac{(-1)^i}{i!} = \frac{1}{k!} \sum_{i=0}^{m-k} \frac{(-1)^i}{i!}$$

□

Definition 1.37. Given $(\Omega, \mathcal{S}, \mathbb{P})$ assume that $H_i \in \mathcal{S}, i \in \mathbb{N}$ such that $\forall i \neq j, H_i \cap H_j = \emptyset$ and $\bigcup_{i=1}^{\infty} H_i = \Omega$. This is a partition of Ω .

Proposition 1.38. Let $\{H_i : i \in \mathbb{N}\}$ be a partition of Ω and assume that $\mathbb{P}(H_i) > 0, \forall i \in \mathbb{N}$. Then $\forall A \in \mathcal{S}$,

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A|H_i)\mathbb{P}(H_i)$$

Proof. $A = \bigcup_{i=1}^{\infty} (A \cap H_i)$ and the $(A \cap H_i)$ are disjoint. Thus

$$\mathbb{P}(A) = \sum_{i=1}^{\infty} \mathbb{P}(A \cap H_i) = \sum_{i=1}^{\infty} \mathbb{P}(A|H_i)\mathbb{P}(H_i)$$

□

Theorem 1.39. (*Bayes Rule*)

Let $\{H_i : i \in \mathbb{N}\}$ be a partition with $\mathbb{P}(H_i) > 0, \forall i \in \mathbb{N}$ and $A \in \mathcal{S}$ with $\mathbb{P}(A) > 0$. Then

$$\mathbb{P}(H_i|A) = \frac{\mathbb{P}(H_i)\mathbb{P}(A|H_i)}{\sum_{j=1}^{\infty} \mathbb{P}(H_j)\mathbb{P}(A|H_j)}$$

Remark 1.40. The intuition for this rule is to think of each H_i as a possible cause of A . Then the left hand side of the rule is determining the likelihood of each H_i actually causing A , if A occurs.

Proof.

$$\mathbb{P}(H_i|A) = \frac{\mathbb{P}(H_i \cap A)}{\mathbb{P}(A)} = \frac{\mathbb{P}(H_i)\mathbb{P}(A|H_i)}{\sum_{j=1}^{\infty} \mathbb{P}(H_j)\mathbb{P}(A|H_j)}$$

□

Example 1.41. Three cars, each of which is coloured on both sides. One car has both sides being red, one has both sides black, the third car has one side red and one side black. Choose a car at random, and show one side of the car at random and observe that it is red. What is the probability that the other side of the car chosen is also red?

Proof. Define H_i to be the event that the i th car is chosen ($i = 1, 2, 3$), A to be the event that the side shown is red. Then we are looking for $\mathbb{P}(H_1|A)$.

$$\mathbb{P}(H_1|A) = \frac{\mathbb{P}(H_1)\mathbb{P}(A|H_1)}{\sum_{j=1}^3 \mathbb{P}(H_j)\mathbb{P}(A|H_j)} = \frac{\frac{1}{3} \cdot 1}{\frac{1}{3} \cdot 1 + 0 + \frac{1}{3} \cdot \frac{1}{2}} = \frac{2}{3}$$

□

Example 1.42. There are $n + 1$ urns marked $0, \dots, n$. Each of these urns contains n balls. The k th urn contains k blue balls and $n - k$ yellow balls. An urn is chosen at random, and m draws are made from it, where the ball drawn is always put back before the next draw. Iff all m draws are blue, then what is the probability that the $m + 1$ draw is also blue.

Proof. Define H_k to be the event that the k th urn is chosen, $\mathbb{P}(H_k) = \frac{1}{k+1}$, $k = 0, 1, \dots, n$. Define A to be the event that the m draws are blue. Define B to be the event that the $m+1$ draw is blue.

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(B \cap A)}{\mathbb{P}(A)} = \frac{\sum_{k=0}^n \mathbb{P}(H_k) \mathbb{P}(B \cap A|H_k)}{\sum_{j=0}^n \mathbb{P}(H_j) \mathbb{P}(A|H_j)} = \frac{\sum_{k=0}^n \frac{1}{n+1} \cdot \left(\frac{k}{n}\right)^{m+1}}{\sum_{j=0}^n \frac{1}{n+1} \left(\frac{j}{n}\right)^m} = \frac{\sum_{k=0}^n k^{m+1}}{n \sum_{j=0}^n j^m}$$

Now consider what happens as $n \rightarrow \infty$. Note the bound

$$\begin{aligned} \sum_{k=1}^n \int_{k-1}^k x^{m+1} dx &\leq \sum_{k=0}^n k^{m+1} \leq \sum_{k=0}^n \int_k^{k+1} x^{m+1} dx \\ \implies \int_1^n x^{m+1} dx &\leq \sum_{k=0}^n k^{m+1} \leq \int_0^{n+1} x^{m+1} dx \\ \implies \frac{n^{m+2} - 1}{m+1} &\leq \sum_{k=0}^n k^{m+1} \leq \frac{(n+1)^{m+2}}{m+2} \\ \implies \sum_{k=0}^n k^{m+1} &= \frac{1}{m+2} n^{m+2} + \mathcal{O}(n^{m+1}) \end{aligned}$$

and similarly

$$\begin{aligned} \sum_{j=0}^n j^m &= \frac{1}{m+1} n^{m+1} + \mathcal{O}(n^m) \\ \implies \frac{\sum_{k=0}^n k^{m+1}}{n \sum_{j=0}^n j^m} &= \frac{\sum_{k=0}^n k^{m+1}}{\sum_{j=0}^n j^m} = \frac{\frac{1}{m+2} n^{m+2} + \mathcal{O}(n^{m+1})}{\frac{1}{m+1} n^{m+1} + \mathcal{O}(n^m)} \end{aligned}$$

as $n \rightarrow \infty$ this goes to

$$\frac{m+1}{m+2}$$

□

1.3 Independence of Events

Definition 1.43. Given $(\Omega, \mathcal{S}, \mathbb{P})$; $\{A_1, \dots, A_n\} \subset \mathcal{S}$. We say that $\{A_1, \dots, A_n\}$ is an independent family (or that they are independent events) if $\forall k = 1, \dots, n, \forall 1 \leq i_1 < i_2 < \dots < i_k \leq n$ we have

$$\mathbb{P}\left(\bigcap_{j=1}^k A_{i_j}\right) = \prod_{j=1}^k \mathbb{P}(A_{i_j})$$

In particular $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

Definition 1.44. Given $(\Omega, \mathcal{S}, \mathbb{P})$ let $\mathcal{E} \subset \mathcal{S}$ be a collection of events. $\mathcal{E}$ is an independent family if every finite subcollection of $\mathcal{E}$ is an independent family.

Definition 1.45. Given $(\Omega, \mathcal{S}, \mathbb{P})$ let $\mathcal{E} \subset \mathcal{S}$ be a collection of events. $\mathcal{E}$ is a pairwise independent family if $\forall A, B \in \mathcal{E}; A \neq B$ we have

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$$

Example 1.46. Toss a fair coin twice. $\Omega = \{HH, HT, TH, TT\}$. Define the following events.

$$E_1 = \{HH, HT\}; \quad E_2 = \{HH, TH\}; \quad E_3 = \{HH, TT\}$$

Every event has $\mathbb{P}(E_i) = \frac{1}{2} \implies \mathbb{P}(E_i \cap E_j) = \frac{1}{4} = \mathbb{P}(E_i)\mathbb{P}(E_j)$. However, $\mathbb{P}(E_1 \cap E_2 \cap E_3) = \frac{1}{2} \neq \frac{1}{8} = \mathbb{P}(E_1)\mathbb{P}(E_2)\mathbb{P}(E_3)$.

Remark 1.47. $A, B \in \mathcal{S}$.

- if $A \subset B$ then A and B cannot be independent unless $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 1$.
- if $A \cap B = \emptyset$ then A and B are not independent unless $\mathbb{P}(A) = 0$ or $\mathbb{P}(B) = 0$
- if A and B are independent, $\mathbb{P}(B) > 0$ then $\mathbb{P}(A|B) = \mathbb{P}(A)$
- A and B are independent $\iff A$ and B^c are independent $\iff A^c$ and B^c are independent $\iff A^c$ and B are independent

Example 1.48. An urn contains k blue balls and $n - k$ yellow balls. 2 draws are made from it, where the ball drawn is put back before the next draw.

Proof. Mark the balls $B_1, \dots, B_k$ (blue) and $B_{k+1}, \dots, B_n$ (yellow).

$$\Omega = \{B_i B_j : 1 \leq i, j \leq n\};$$

$$E_1 = \{B_i B_j : 1 \leq i \leq k, 1 \leq j \leq n\}; E_2 = \{B_i B_j : 1 \leq i \leq n, k+1 \leq j \leq n\}$$

$$E_1 \cap E_2 = \{B_i B_j : 1 \leq i \leq k, k+1 \leq j \leq n\}$$

Now we have

$$\mathbb{P}(E_1) = \frac{|E_1|}{|\Omega|} = \frac{kn}{n^2} = \frac{k}{n}$$

$$\mathbb{P}(E_2) = \frac{|E_2|}{|\Omega|} = \frac{(n-k)n}{n^2} = \frac{n-k}{n}$$

$$\mathbb{P}(E_1 \cap E_2) = \frac{|E_1 \cap E_2|}{|\Omega|} = \frac{(n-k)k}{n^2} = \mathbb{P}(E_1)\mathbb{P}(E_2)$$

Thus E_1, E_2 are independent □

Example 1.49. An urn contains k blue balls and $n - k$ yellow balls. 2 draws are made from it, where the ball drawn is removed before the next draw.

Proof. Mark the balls $B_1, \dots, B_k$ (blue) and $B_{k+1}, \dots, B_n$ (yellow).

$$\Omega = \{B_i B_j : 1 \leq i, j \leq n; j \neq i\};$$

$$E_1 = \{B_i B_j : 1 \leq i \leq k, 1 \leq j \leq n; j \neq i\}; E_2 = \{B_i B_j : 1 \leq i \leq n, k+1 \leq j \leq n; j \neq i\}$$

$$E_1 \cap E_2 = \{B_i B_j : 1 \leq i \leq k, k+1 \leq j \leq n\}$$

Now we have

$$\mathbb{P}(E_1) = \frac{|E_1|}{|\Omega|} = \frac{k(n-1)}{n(n-1)} = \frac{k}{n}$$

$$\mathbb{P}(E_2) = \frac{|E_2|}{|\Omega|} = \frac{(n-k)k + (n-k)(n-k-1)}{n(n-1)} = \frac{n-k}{n}$$

$$\mathbb{P}(E_1 \cap E_2) = \frac{|E_1 \cap E_2|}{|\Omega|} = \frac{(n-k)k}{n(n-1)} > \mathbb{P}(E_1)\mathbb{P}(E_2)$$

So now E_1, E_2 are not independent. □

2 Discrete and Continuous Random Variables

2.1 Distributions

Definition 2.1. If Ω is finite – i.e. $\Omega = \{\omega_1, \dots, \omega_n\}$ – or countable – $\Omega = \{\omega_i : i \in \mathbb{N}\}$ – and $\mathcal{S} = 2^\Omega$ then it suffices to assign on $\{\omega\}, \omega \in \Omega$ then $\forall A \in \mathcal{S}, \mathbb{P}(A) = \sum_{\omega \in A} \mathbb{P}(\{\omega\})$. In this case, $(\Omega, \mathcal{S}, \mathbb{P})$ is a discrete probability space, $\mathbb{P}$ a discrete probability or a discrete distribution.

Definition 2.2. The function $f : \Omega \rightarrow [0, 1]$ given by $f(\omega) = \mathbb{P}(\{\omega\}), \forall \omega \in \Omega$ is the probability mass function (PMF) of $\mathbb{P}$.

Proposition 2.3. Given discrete Ω and $\mathcal{S} = 2^\Omega$ we have $f : \Omega \rightarrow [0, 1]$ is the PMF of some $\mathbb{P}$ on $(\Omega, \mathcal{S})$ if and only if $\sum_{\omega \in \Omega} f(\omega) = 1$.

Proof. The forward direction is obvious, the reverse comes from defining $\mathbb{P}(\{\omega\}) = f(\omega)$. □

The following are examples of discrete distributions – $\mathcal{S} = 2^\Omega$

Example 2.4. $\Omega = \{\omega_1, \dots, \omega_n\}$ then $\mathbb{P}$ is the discrete uniform distribution if the PMF is $f(\omega_i) = \frac{1}{n}, i = 1, \dots, n$

Example 2.5. $\omega_0 \in \Omega$. $\mathbb{P}$ is the Dirac distribution at ω_0 if PMF is $f(\omega_0) = 1, f \equiv 0$ on $\Omega - \{\omega_0\}$

Example 2.6. $\Omega = \{0, 1\}; p \in (0, 1)$. $\mathbb{P}$ is the Bernoulli distribution denoted by $B(p)$ if the PMF is $f(1) = p, f(0) = 1 - p$.

$B(p)$ is used to model trials with only two outcomes. (success/failure)

Example 2.7. $\Omega = \{0, 1, \dots, n\}, p \in (0, 1)$. $\mathbb{P}$ is the binomial distribution denoted by $B(n, p)$ if the PMF is $F(k) = \binom{n}{k} p^k (1 - p)^{n-k}; k = 0, 1, \dots, n$. We have $\sum_{k=0}^n f(k) = \sum_{k=0}^n \binom{n}{k} p^k (1 - p)^{n-k} = (p + 1 - p)^n = 1$ so f is a PMF.

$B(n, p)$ models the number of successes in n independent Bernoulli trials. $\mathbb{P}(\text{success in } k \text{ out of } n \text{ trials}) = \binom{n}{k} p^k (1 - p)^{n-k} \rightarrow B(n, p)$

Example 2.8. $\Omega = \{0\} \cup \mathbb{N}; \lambda > 0$. $\mathbb{P}$ is the Poisson distribution denoted by $P(\lambda)$. The PMF is $f(k) = e^{-\lambda} \frac{\lambda^k}{k!}, k = 0, 1, 2, \dots$. We have

$$\sum_{k=0}^{\infty} f(k) = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1$$

models the number of occurrences of rare events.

Example 2.9. Study the number of Pokémon you find in a day. Assume that the likelihood of you finding a Pokémon in a day is $\lambda > 0$. Equally divide the day into n time intervals with n large. Then within each time interval your chance of finding one Pokémon is $\frac{\lambda}{n}$.

The number of Pokémon found in a day is the number of successes you have in a day $\approx B(n, \frac{\lambda}{n})$.

PMF of $B(n, \frac{\lambda}{n})$:

$$\begin{aligned} f(k) &= \binom{n}{k} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k} = \frac{n!}{k!(n-k)!} \cdot \frac{\lambda^k}{n^k} \cdot \frac{(n-\lambda)^{n-k}}{n^{n-k}} \\ &= \frac{\lambda^k}{k!} \cdot \frac{n(n-1) \cdots (n-k+1)(n-\lambda)^k}{n^n(n-\lambda)^k} \\ &= \frac{\lambda^k}{k!} \left(1 - \frac{\lambda}{n}\right)^n \cdot \left(\frac{n}{n-\lambda}\right) \cdot \left(\frac{n-1}{n-\lambda}\right) \cdots \left(\frac{n-k+1}{n-\lambda}\right) \xrightarrow{n \rightarrow \infty} \frac{\lambda^k}{k!} e^{-\lambda} = P(\lambda) \end{aligned}$$

So $P(\lambda)$ is the limit of $B(n, \frac{\lambda}{n})$.

Definition 2.10. Consider the probability $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{Q})$ – i.e. $\mathbb{Q}$ is a probability on the space.

The distribution function (DF) F of $\mathbb{Q}$ is defined as $F : \mathbb{R} \rightarrow [0, 1]$ and $F(x) = \mathbb{Q}((-\infty, x]), \forall x \in \mathbb{R}$.

Properties of F :

- F is nondecreasing on $\mathbb{R}$
- $\lim_{x \rightarrow -\infty} F(x) = 0; \lim_{x \rightarrow \infty} F(x) = 1$ – this is because $\{(-\infty, -n] : n \in \mathbb{N}\}$ is a monotonic (non increasing) sequence, and $\bigcap_{n=1}^{\infty} (-\infty, -n] = \emptyset$. Similar for the other limit.

- F is continuous from the right i.e., $\forall a \in \mathbb{R}, \lim_{x \rightarrow a^+} F(x) = F(a)$

Remark 2.11. Fix $a \in \mathbb{R}$, and let $\{x_n\}$ such that $x_n \rightarrow a^+$ as $n \rightarrow \infty$.

$$\begin{aligned} (-\infty, x_n] \downarrow \bigcap_{n=1}^{\infty} (-\infty, x_n] &= (-\infty, a] \\ \implies \lim_{n \rightarrow \infty} \mathbb{Q}((-\infty, x_n]) &= \mathbb{Q}((-\infty, a]) \implies \lim_{n \rightarrow \infty} F(x_n) = F(a) \end{aligned}$$

Remark 2.12. F is not necessarily left continuous.

Proposition 2.13. If $F : \mathbb{R} \rightarrow [0, 1]$ satisfies (i), (ii), (iii) from the above then F is the DF for some probability $\mathbb{Q}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Proof. Proof is omitted here. The idea is take F that satisfies the three properties and define $\mathbb{Q} : (a, b] \mapsto F(b) - F(a)$. Argue that $\mathbb{Q}$ is a probability on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Check that $\mathbb{Q} \geq 0$, $\mathbb{Q}(\mathbb{R}) = 1$, $\mathbb{Q}$ is countably additive. $\square$

Remark 2.14. Many discrete distributions can be thought of as distributions on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then the DF of a discrete distribution is a jump function/step function/piece-wise continuous function.

Example 2.15. Binomial distribution $B(n, p), \Omega\{0, 1, \dots, n\} \subset \mathbb{R}, \mathcal{S} = 2^\Omega \subset \mathcal{B}(\mathbb{R})$. The DF of $B(n, p)$:

$$F(x) = \mathbb{P}((-\infty, x]) = \sum_{k \in \{0, 1, \dots, n\}, k \leq x} f(k) = \sum_{k \in \{0, \dots, n\}, k \leq x} \binom{n}{k} p^k (1-p)^{n-k}$$

where f is the PMF of $B(n, p)$.

- if $x < 0$ then $F(x) = 0$.
- if $0 \leq x < 1$ then $F(x) = (1-p)^n$
- if $1 \leq x < 2$ then $F(x) = (1-p)^n + np(1-p)^{n-1}$

Example 2.16. Poisson distribution $P(\lambda), \Omega\{0, 1, 2, \dots\} \subset \mathbb{R}, \mathcal{S} = 2^\Omega \subset \mathcal{B}(\mathbb{R})$. The DF of $B(n, p)$:

$$F(x) = \mathbb{P}((-\infty, x]) = \sum_{k \in \mathbb{N}, k \leq x} f(k) = \sum_{k \in \mathbb{N}, k \leq x} e^{-\lambda} \frac{\lambda^k}{k!}$$

where f is the PMF of $P(\lambda)$.

- if $x < 0$ then $F(x) = 0$.

- if $0 \leq x < 1$ then $F(x) = e^{-\lambda}$
- if $1 \leq x < 2$ then $F(x) = e^{-\lambda} + \lambda e^{-\lambda}$
- $\forall x \in \mathbb{R}, F(x) < 1$ because there will always be values greater than it, and the jumps will get smaller and smaller

Remark 2.17. PMF is more useful for discrete distributions.

Example 2.18. Define:

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x \in [0, 1] \\ 1 & \text{if } x \geq 1 \end{cases} \quad (2.1)$$

F is a DF for some $\mathbb{Q}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

- $\mathbb{Q}((-\infty, x]) = F(x)$. Check that $\mathbb{Q}(B) = 0$ if $B \subset (-\infty, 0]$.
- $\forall$ intervals $I \subset [0, 1]$ then $\mathbb{Q}(I) = \text{length of } I$
- $\forall A \subset \mathbb{R}$ such that $[0, 1] \subset A, \mathbb{Q}(A) = 1$

$\mathbb{Q}$ is the uniform distribution on $[0, 1] : ([0, 1], \mathcal{B}([0, 1]), \mathbb{Q})$

Definition 2.19. Given $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{Q})$ and the DF of $\mathbb{Q}$, say F , $\mathbb{Q}$ is a continuous distribution if F is absolutely continuous, i.e., $\exists f : \mathbb{R} \rightarrow [0, \infty)$ such that $\forall x \in \mathbb{R}$

$$F(x) = \int_{-\infty}^x f(t)dt$$

Then f is the probability density function of $\mathbb{Q}$.

Remark 2.20. If $\mathbb{Q}$ is a continuous distribution with DF $F(x)$ and PDF $f(t)$

- $\int_{-\infty}^{\infty} f(t)dt = \lim_{x \rightarrow \infty} F(x) = 1$
- $\forall a \leq b, \mathbb{Q}((a, b]) = F(b) - F(a) = \int_a^b f(t)dt$
- F is continuous everywhere, i.e. $\forall a \in \mathbb{R}, \lim_{x \rightarrow a} F(x) = F(a)$. Also F is differentiable almost everywhere, i.e. $F'(x)$ exists for almost every $x \in \mathbb{R}$ and $F'(x) = f(x)$ for almost every x as well. However, if f is continuous at x then $F'(x) = f(x)$

Example 2.21. Consider again the DF

$$F(x) = \begin{cases} 0 & \text{if } x \leq 0 \\ x & \text{if } x \in [0, 1] \\ 1 & \text{if } x \geq 1 \end{cases} \quad (2.2)$$

Then we have the PDF

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \in (0, 1) \\ 0 & \text{if } x > 1 \end{cases} \quad (2.3)$$

Example 2.22. Normal (Gaussian) distribution denoted by $N(0, 1)$ is a continuous distribution. $N(0, 1)$ has DF, PDF

$$F(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\frac{t^2}{2}} dt; \quad f(t) = \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}}$$

Remark 2.23. PDF does not exist for discrete distributions.

Remark 2.24. There are distributions in between the discrete and continuous types, e.g.

$$F(x) = \begin{cases} 0 & \text{if } x < 0 \\ \frac{1}{4} + \frac{1}{2}x & \text{if } 0 \leq x < \frac{1}{2} \\ \frac{3}{4} & \text{if } \frac{1}{2} \leq x < 1 \\ 1 - \frac{1}{8}e^{1-x} & \text{if } x \geq 1 \end{cases} \quad (2.4)$$

Remark 2.25. Exercise: Find $a \in (0, 1)$, continuous F_1 , discrete F_2 such that $F(x) = aF_1(x) + (1 - a)F_2(x)$.

Remark 2.26. There are DF $F(x)$ such that $F(x)$ is continuous but $F(x)$ is not absolutely continuous.

Example 2.27. Cantor function.

Remark 2.28. If $\mathbb{Q}$ is a continuous distribution then $\forall x \in \mathbb{R}, \mathbb{Q}(\{x\}) = 0$.

Remark 2.29. View $f(x)$ as the probability near point x .

$$\mathbb{Q}(x - \epsilon, x + \epsilon) = \int_{x-\epsilon}^{x+\epsilon} f(t) dt \approx f(x)2\epsilon$$

2.2 Random Variables

2.2.1 Discrete Random Variables

Definition 2.30. Let Ω be a sample space, $\mathcal{S}$ a σ -field on Ω . A function $X : \Omega \rightarrow \mathbb{R}$ is called a random variable (RV) if

$$\forall B \in \mathcal{B}(\mathbb{R}), X^{-1}(B) := \{\omega \in \Omega : X(\omega) \in B\} \in \mathcal{S}$$

Remark 2.31. Properties of a random variable:

- if Ω is discrete and $\mathcal{S} = 2^\Omega$ then any function $X : \Omega \rightarrow \mathbb{R}$ is a RV

- if $\Omega = \mathbb{R}, \mathcal{S} = \mathcal{B}(\mathbb{R})$ then $\exists$ a function $X : \mathbb{R} \rightarrow \mathbb{R}$ that is not a RV, e.g., for $A \subset \mathbb{R}$ define $X = \chi_A$. Then we have X is a RV $\iff A \in \mathcal{B}(\mathbb{R})$. Note, this definition is:

$$X(x) = \chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{otherwise} \end{cases} \quad (2.5)$$

Now $\forall B \in \mathcal{B}(\mathbb{R})$

$$X^{-1}(B) = \begin{cases} \emptyset & \text{if } 0, 1 \notin B \\ A^c & \text{if } 0 \in B, 1 \notin B \\ A & \text{if } 1 \in B, 0 \notin B \\ \mathbb{R} & \text{if } 0, 1 \in B \end{cases} \quad (2.6)$$

Example 2.32. Examples of a RV

- roll a pair of dice. X = sum of the face values is a RV
- randomly choose two points a, b from $[0, 1]$, $X = |a - b|$
- looking for Pokémon in a day, X = the time that you find the fifth Pokémon

Remark 2.33. Exercise: Verify that for every function $X : \Omega \rightarrow \mathbb{R}, A, B \subset \mathbb{R}, \{B_i : i \in \mathbb{N}\} \subset \mathbb{R}$

- $X^{-1}(B^c) = (X^{-1}(B))^c$
- $X^{-1}(A \cap B) = (X^{-1}(A)) \cap (X^{-1}(B)); \quad X^{-1}(A \cup B) = (X^{-1}(A)) \cup (X^{-1}(B))$
- $X^{-1}(\bigcap_{i=1}^{\infty} B_i) = \bigcap_{i=1}^{\infty} (X^{-1}(B_i)); \quad X^{-1}(\bigcup_{i=1}^{\infty} B_i) = \bigcup_{i=1}^{\infty} (X^{-1}(B_i))$

Theorem 2.34. Given $(\Omega, \mathcal{S})$ and $X : \Omega \rightarrow \mathbb{R}$, X is a RV if and only if $\forall x \in \mathbb{R}$

$$\{X \leq x\} = \{\omega \in \Omega : X(\omega) \leq x\} \in \mathcal{S}$$

Proof. ($\implies$) by definition of a RV.

($\impliedby$) Define $\mathcal{D} = \{B \subset \mathbb{R} : X^{-1}(B) \in \mathcal{S}\}$. Show that $\mathcal{D}$ is a σ -field.

- $X^{-1}(\mathbb{R}) = \Omega \in \mathcal{S}$
- if $B \in \mathcal{D}$ then $X^{-1}(B) \in \mathcal{S} \implies X^{-1}(B^c) = (X^{-1}(B))^c \in \mathcal{S} \implies B^c \in \mathcal{D}$
- if $B_i \in \mathcal{D}, i \in \mathbb{N}$ then $\bigcup_{i=1}^{\infty} B_i \in \mathcal{D}$

Now we have that $\mathcal{D}$ is a σ -field.

We know that

$$\forall x \in \mathbb{R}, (-\infty, x] \in \mathcal{D} \implies \{(-\infty, x] : x \in \mathbb{R}\} \subset \mathcal{D}$$

It suffices to show that $\sigma(\{(-\infty, x] : x \in \mathbb{R}\}) = \mathcal{B}(\mathbb{R}) = \sigma(\{(a, b) : a < b; a, b \in \mathbb{R}\})$.

Since $(-\infty, x] \in \mathcal{B}(\mathbb{R})$, $\text{LHS} \subset \text{RHS}$.

$\forall a, b \in \mathbb{R}, (a, b] = (-\infty, b] - (-\infty, a]$ and $(a, b) = \bigcup_{n=1}^{\infty} (a, b - \frac{1}{n}]$ so we have equality. $\square$

Corollary 2.35. *IF X is a RV on $(\Omega, \mathcal{S})$ then*

$$aX + b (a, b \in \mathbb{R}, a \neq 0); \quad |X|, \quad X^2; \quad \frac{1}{X} (X \neq 0); \quad \sqrt{X} (X \geq 0); \quad e^X; \quad \log X (X > 0)$$

are all RV.

Proof. First, $aX + b$. $\forall x \in \mathbb{R}$,

$$\begin{aligned} \{aX + b \leq x\} &= \{\omega \in \Omega : aX + b \leq x\} \\ &= \{\omega \in \Omega : aX \leq x - b\} \end{aligned}$$

WLOG $a < 0$

$$= \{\omega \in \Omega : X \geq \frac{x - b}{a}\} = \{X \geq \frac{x - b}{a}\} \in \mathcal{S}$$

Now, $|X|$. $\forall x \in [0, \infty$

$$\{|X| \leq x\} = \begin{cases} \{-x \leq X \leq x\} = X^{-1}([-x, x]) \in \mathcal{S} : & \text{if } x \geq 0 \\ \{|X| \leq x\} = \emptyset \in \mathcal{S} : & \text{otherwise} \end{cases} \quad (2.7)$$

Others are similar. $\square$

Proposition 2.36. *Given $(\Omega, \mathcal{S})$ and a RV $X : \Omega \rightarrow \mathbb{R}$, let $g : \mathbb{R} \rightarrow \mathbb{R}$ - a RV - such that $\forall B \in \mathcal{B}(\mathbb{R}), g^{-1}(B) \in \mathcal{B}(\mathbb{R})$. Then $g \circ X : \Omega \rightarrow \mathbb{R}$ is a RV.*

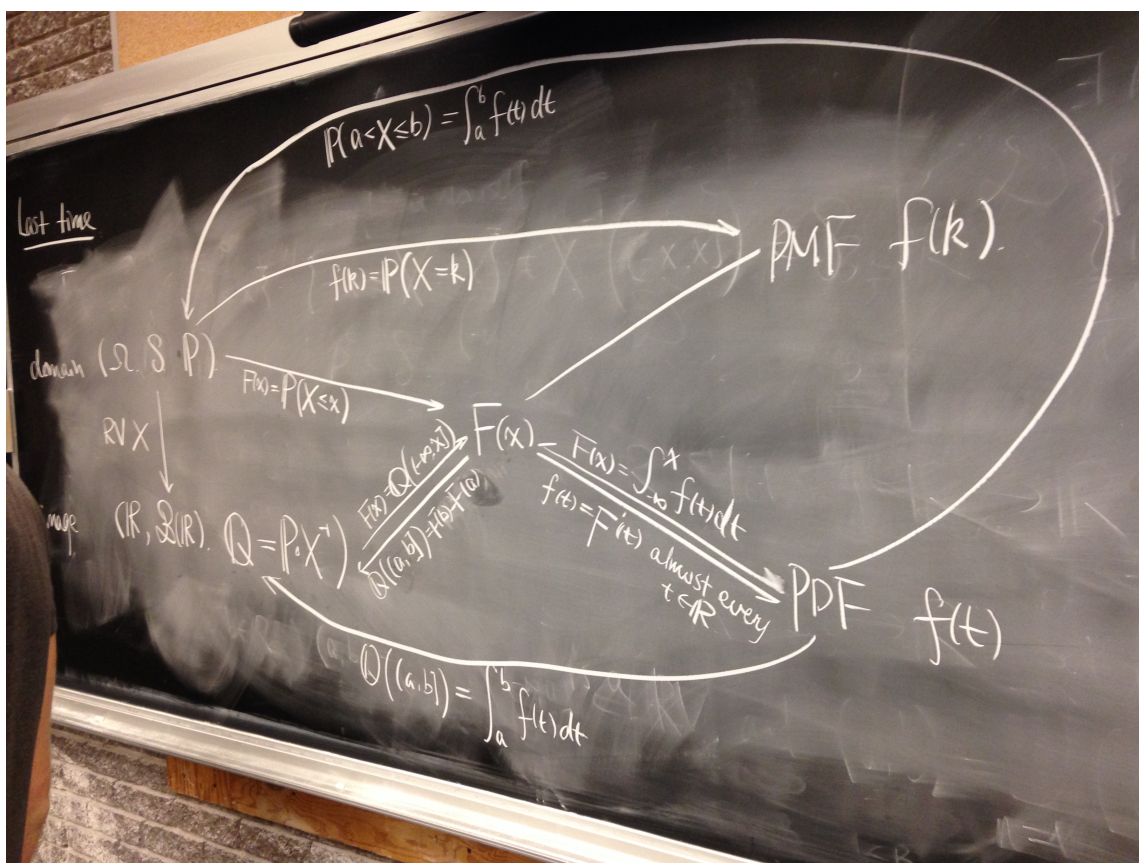
Proof. Exercise. $\square$

Remark 2.37. $\exists A \subset \mathbb{R}$ such that $A \notin \mathcal{B}(\mathbb{R})$. If $g = \chi_A$ then $g \circ X$ is not a RV.

Definition 2.38. Given $(\Omega, \mathcal{S}, \mathbb{P})$ let $X : \Omega \rightarrow \mathbb{R}$ be a RV. Then define the following set function on $\mathcal{B}(\mathbb{R})$

$$\forall B \in \mathcal{B}(\mathbb{R}), \mathbb{Q}(B) := \mathbb{P}(X^{-1}(B))$$

$\mathbb{Q}$ is the distribution of RV X . If F is the DF of $\mathbb{Q}$, i.e. $F(x) = \mathbb{Q}((-\infty, x])$, $\forall x \in \mathbb{R}$. Then F is also called the DF of X . $F(x) = \mathbb{P}(X \leq x)$.



Theorem 2.39. If $\mathbb{Q}$ is a probability distribution on $(\mathbb{R}, \mathcal{B}(\mathbb{R})) \iff \exists (\Omega, \mathcal{S}, \mathbb{P})$ and a RV $X : \Omega \rightarrow \mathbb{R}$ such that $\mathbb{Q} = \mathbb{P} \circ X^{-1}$.

Proof. ($\Leftarrow$) Check the definition.

- $\forall B \in \mathcal{B}(\mathbb{R}), \mathbb{Q}(B) = \mathbb{P}(X^{-1}(B)) \geq 0$
- $\mathbb{Q}(\mathbb{R}) = \mathbb{P}(X^{-1}(\mathbb{R})) = \mathbb{P}(\Omega) = 1$
- $\forall B_i \in \mathcal{B}(\mathbb{R}), i \in \mathbb{N}$ with the B_i 's disjoint then

$$\begin{aligned} \mathbb{Q}\left(\bigcup_{i=1}^{\infty} B_i\right) &= \mathbb{P}\left(X^{-1}\left(\bigcup_{i=1}^{\infty} B_i\right)\right) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} X^{-1}(B_i)\right) \\ &= \sum_{i=1}^{\infty} \mathbb{P}(X^{-1}(B_i)) = \sum_{i=1}^{\infty} \mathbb{Q}(B_i) \end{aligned}$$

($\implies$) Exercis.

Take $(\Omega, \mathcal{S}, \mathbb{P}) = (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{Q})$ Define $X = Id, X(x) = x, \forall x \in \mathbb{R}$.

Reference: Skorohod's Representation □

Definition 2.40. We say that X is a discrete uniform/Dirac/Bernoulli/Binomial/Poisson/.../uniform on $[0,1]$ /normal... RV if the distribution $\mathbb{Q}$ of X discrete uniform/Dirac/Bernoulli/Binomial/Poisson/.../uniform on $[0,1]$ /normal...

Definition 2.41. If the distribution $\mathbb{Q}$ of X is a continuous(discrete) distribution with PDF $f(x)$ (PMF $f(k)$) then f is also called the PDF(PMF) of X .

Definition 2.42. A transformation of a RV is the composition of a RV, X , with a Borel function, g , i.e., $g \circ X$. The distribution $\mathbb{Q}_{g \circ X}$ of $g \circ X$ is given by

$$B \in \mathcal{B}(\mathbb{R}) \mapsto \mathbb{Q}_{g \circ X}(B) = \mathbb{P}((g \circ X)^{-1}(B)) = \mathbb{P}(X \in g^{-1}(B)) = \mathbb{Q}_X(g^{-1}(B))$$

Remark 2.43. If X is a discrete RV, i.e., $X \in \{x_n : n \in \mathbb{N}\}$ (X takes values in $\{x_n\}$) then $\forall B \in \mathcal{B}(\mathbb{R})$

$$\mathbb{Q}_{g \circ X}(B) = \mathbb{P}(X \in g^{-1}(B)) = \sum_{\{n \in \mathbb{N} : x_n \in g^{-1}(B)\}} \mathbb{P}(X = x_n) = \sum_{\{n \in \mathbb{N} : x_n \in g^{-1}(B)\}} f_X(x_n)$$

f_X is the PMF of X .

Remark 2.44. Note that if X is discrete, then $g \circ X$ is also a discrete RV and $g \circ X \in \{g(x_n) : n \in \mathbb{N}\}$. If $f_{g \circ X}$ is the PMF of $g \circ X$ then

$$f_{g \circ X}(g(x_n)) = \mathbb{Q}_{g \circ X}(\{g(x_n)\}) = \sum_{\{k \in \mathbb{N} : g(x_k) = g(x_n)\}} f_X(x_k)$$

Example 2.45. X is a $P(\lambda)$ RV. $Y = e^X$ is a discrete RV. $X \in \{0, 1, 2, 3, \dots\}; Y \in \{1, e, e^1, e^2, e^3, \dots\}$.

$$f_Y(e^k) = \sum_{\{n \in \mathbb{N} : e^n = e^k\}} f_X(n) = f_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

Example 2.46. X is a $B(4, p)$ RV. $X \in \{0, 1, 2, 3, 4\}; Y = (X - 2)^2 \in \{4, 1, 0\}$. $g(x) = (x - 2)^2$

$$f_Y(4) = f_X(0) + f_X(4) = \binom{4}{0} p^0 (1-p)^4 + \binom{4}{4} p^4 (1-p)^0$$

$$f_Y(1) = f_X(1) + f_X(3) = \binom{4}{1} p^1 (1-p)^3 + \binom{4}{3} p^3 (1-p)^1$$

$$f_Y(0) = f_X(2) = \binom{4}{2} p^2 (1-p)^2$$

2.2.2 Expectation of a Discrete Random Variable

Definition 2.47. Assume that X is a discrete RV on $(\Omega, \mathcal{S}, \mathbb{P})$ and $X \in \{x_n : n \in \mathbb{N}\}$ with PMF of X being $f_X(x_n) = \mathbb{P}(X = x_n), \forall n \in \mathbb{N}$. If

$$\sum_{n=1}^{\infty} |x_n| f_X(x_n) = \sum_{n=1}^{\infty} |x_n| \mathbb{P}(X = x_n) < \infty$$

then we say that the expectation of X exists and it is defined to be

$$\mathbb{E}[X] = \sum_{n=1}^{\infty} x_n f_X(x_n) = \sum_{n=1}^{\infty} x_n \mathbb{P}(X = x_n)$$

If $\sum_{n=1}^{\infty} |x_n| \mathbb{P}(X = x_n) = \infty$ then we say that the expectation does not exist.

Remark 2.48. X is a RV with PMF

$$f_X \left((-1)^{j+1} \frac{3^j}{j} \right) = \frac{1}{3^j}, j = 1, 2, 3, \dots$$

Check that it is a PMF, i.e., $\sum_{j=1}^{\infty} \frac{2}{3^j} = 1$. Now,

$$\sum_{j=1}^{\infty} x_j \frac{2}{3^j} = 2 \sum_{j=1}^{\infty} (-1)^{j+1} \frac{3^j}{j} \frac{2}{3^j} = 2 \sum_{j=1}^{\infty} (-1)^{j+1} \frac{1}{j} = 2 \ln 2$$

Note $\ln(1+x) = \sum_{n=1}^{\infty} (-1)^{n+1} \frac{x^n}{n}$ when $0 < x \leq 1$. However:

$$\sum_{j=1}^{\infty} \frac{2}{j} = \infty$$

by comparison with the integral $\int_1^{\infty} \frac{1}{x} dx$. Thus, $\mathbb{E}[X]$ does not exist.

Remark 2.49. • $\mathbb{E}[X]$ exists if and only if $\mathbb{E}[X] < \infty$

- If $X \geq 0$ then if $\mathbb{E}[X]$ exists, $\mathbb{E}[X] = \mathbb{E}[|X|]$
- If $\mathbb{E}[X]$ exists then $|\mathbb{E}[X]| \leq \mathbb{E}[|X|]$
- If X is a finite RV then $\mathbb{E}[X], \mathbb{E}[|X|]$ always exists

The expectation is the "average" of X or "mean" of X .

Example 2.50. If X is the discrete uniform RV on $\{x_1, \dots, x_n\}$ then

$$\mathbb{E}[X] = \sum_{i=1}^n x_i \mathbb{P}(X = x_i) = \frac{\sum_{i=1}^n x_i}{n}$$

This is the average.

Example 2.51. If X is the Bernoulli RV ($p \in (0, 1)$), $X \in \{0, 1\}$ then

$$\mathbb{E}[X] = 0(1 - p) + 1 \cdot p = p$$

Example 2.52. If X is $B(n, p)$, $X \in \{0, \dots, n\}$ then

$$\begin{aligned} \mathbb{E}[X] &= \sum_{k=0}^n k \mathbb{P}(X = k) = \sum_{k=0}^n k \binom{n}{k} p^k (1 - p)^{n-k} \\ &= np \sum_{k=1}^n \frac{(n-1)!}{(k-1)!(n-1-(k-1))!} p^{k-1} (1-p)^{n-1-(k-1)} \end{aligned}$$

Now setting $l = k - 1$

$$= np \sum_{l=0}^{n-1} \binom{n-1}{l} p^l (1-p)^{n-1-l} = np$$

Since we have that $\binom{n-1}{l} p^l (1-p)^{n-1-l}$ is the PMF of $B(n-1, p)$.

Example 2.53. If X is $P(\lambda)$, $X \in \{0, 1, 2, \dots\}$, $X \geq 0$ then

$$\begin{aligned} \mathbb{E}[|X|] &= \mathbb{E}[X] = \sum_{k=0}^{\infty} k \mathbb{P}(X = k) = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda} \\ &= \lambda \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} \\ &= \lambda \sum_{l=0}^{\infty} \frac{\lambda^l}{l!} e^{-\lambda} = \lambda \end{aligned}$$

Since $\frac{\lambda^l}{l!} e^{-\lambda}$ is the PMF of $P(\lambda)$.

Remark 2.54. $A \in \mathcal{S}$, $X = \chi_A$, $X \in \{0, 1\}$. Then

$$\mathbb{E}[X] = 1\mathbb{P}(X = 1) = \mathbb{P}(A)$$

Theorem 2.55. $X : \Omega \rightarrow \mathbb{R}$ is a discrete RV, $X \in \{x_n : n \in \mathbb{N}\}$, $g : \mathbb{R} \rightarrow \mathbb{R}$ Borel, $Y = g \circ X$ is a discrete RV. Then

$$\mathbb{E}[X] \text{ exists} \iff \sum_{n=1}^{\infty} |g(x_n)| \mathbb{P}(X = x_n) < \infty$$

and in this case

$$\mathbb{E}[g \circ X] = \mathbb{E}[Y] = \sum_{n=1}^{\infty} g(x_n) \mathbb{P}(X = x_n)$$

Proof. $Y \in \{y_k : k \in \mathbb{N}\} = \{g(x_n) : n \in \mathbb{N}\}.$

$$\begin{aligned}\mathbb{E}[|Y|] &= \sum_{k=1}^{\infty} |y_k| \mathbb{P}(Y = y_k) = \sum_{k=1}^{\infty} |y_k| \sum_{\{n: g(x_n)=y_k\}} \mathbb{P}(X = x_n) \\ &= \sum_{k=1}^{\infty} \sum_{\{n: g(x_n)=y_k\}} |g(x_n)| \mathbb{P}(X = x_n) = \sum_{n=1}^{\infty} |g(x_n)| \mathbb{P}(X = x_n)\end{aligned}$$

since $\mathbb{P}(Y = y_k) = \mathbb{P}(g \circ X = y_k) = \mathbb{P}(X \in g^{-1}(\{y_k\}))$ □

Corollary 2.56. *If $\mathbb{E}[X]$ exists then $\forall a, b, \in \mathbb{R}$, $\mathbb{E}[aX + b]$ exists and*

$$a\mathbb{E}[X] + b$$

In particular, if $\mathbb{E}[X] = m$

$$\mathbb{E}[X - m] = 0$$

Proof. $g(x) = ax + b$, $X \in \{x_n : n \in \mathbb{N}\}.$

$$\sum_n |ax + b| \mathbb{P}(X = x_n) \leq \sum_n (|a| \cdot |x_n| + |b|) \mathbb{P}(X = x_n) = |a| \sum_n |x_n| \mathbb{P}(X = x_n) + |b| < \infty$$

So $\mathbb{E}[aX + b]$ exists.

$$\mathbb{E}[aX + b] = \sum_n (ax_n + b) \mathbb{P}(X = x_n) = a\mathbb{E}[X] + b$$

□

Example 2.57. Let X be a $B(n, p)$ RV. Compute $\mathbb{E}\left[\frac{1}{x+1}\right] \cdot g(x) = \frac{1}{x+1}.$

$$\mathbb{E}\left[\frac{1}{x+1}\right] = \sum_{k=0}^n \frac{1}{k+1} \binom{n}{k} p^k (1-p)^{n-k}$$

Exercise: show this is equal to

$$\frac{1 - (1-p)^{n+1}}{(n+1)p}$$

Note that $\frac{1}{k+1} \binom{n}{k} = \frac{n!}{(k+1)!(n-k)!}$

Definition 2.58. Let X be a RV for $k \in \{0, 1, 2, \dots\}$. If $\mathbb{E}[X^k]$ exists then we can find $\mathbb{E}[X^k]$ is called the k th moment of X , M_k .

$$\mathbb{E}[X^0] = 1$$

$$\mathbb{E}[X^1] = \mathbb{E}[X]$$

Remark 2.59. X is a discrete RV. $X \in \{x_n : n \in \mathbb{N}\}$. If $\sum_n |x_n^k| \mathbb{P}(X = x_n) < \infty$ then $M_k = \sum_n x_n^k \mathbb{P}(X = x_n)$.

Remark 2.60. If I know all the moments of X then I know completely the distribution of X .

Example 2.61. If $X \in \{x_1, \dots, x_n\}$ denote $p_n = \mathbb{P}(X = x_n)$. I know M_k for all the k s so we have the system of linear equations

$$\begin{aligned} M_1 &= \mathbb{E}[X] = \sum_{j=1}^n x_j p_j \\ M_2 &= \mathbb{E}[X^2] = \sum_{j=1}^n x_j^2 p_j \\ &\dots \\ M_n &= \mathbb{E}[X^n] = \sum_{j=1}^n x_j^n p_j \end{aligned}$$

which can be solved for each p_i .

Definition 2.62. X is a RV with $\mathbb{E}[X^2] < \infty$. The variance of X denoted by $\text{Var}(X)$ is defined by

$$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2 - 2X\mathbb{E}[X] + (\mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$$

Remark 2.63. $\text{Var}(X) \geq 0$. Variance measures "how far/much" the distribution of X from $\mathbb{E}[X]$.

Remark 2.64. The variance of X is 0 if and only if X is a Dirac RV.

Example 2.65. $X \in \{x_1, \dots, x_n\}$ a discrete uniform RV.

$$M_2 = \mathbb{E}[X^2] = \sum_{j=1}^n x_j^2 \frac{1}{n} = \frac{1}{n} \sum_{j=1}^n x_j^2; \quad \text{Var}(X) = \frac{\sum_{j=1}^n x_j^2}{n} - \left(\frac{\sum_{j=1}^n x_j}{n} \right)^2$$

Example 2.66. X is the $B(p)$ RV, $X \in \{0, 1\}$

$$\mathbb{E}[X^2] = 0 \cdot (1 - p) + p; \quad \text{Var}(x) = p - p^2$$

Example 2.67. X is a $B(n, p)$ RV.

$$\mathbb{E}[X^2] = \sum_k = 0^n k^2 \binom{n}{k} p^k (1 - p)^{n-k} = np \sum_{k=1}^n k \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} (1-p)^{n-k}$$

now set $l = k - 1$

$$\mathbb{E}[X^2] = np \sum_{l=0}^{n-1} (l+1) \binom{n-1}{l} p^l (1-p)^{n-1-l}$$

if Y is $B(n-1, p)$ RV then $\mathbb{E}[Y+1] = \sum_{l=0}^{n-1} (l+1) \binom{n-1}{l} p^l (1-p)^{n-1-l} = \mathbb{E}[Y] + 1$ so we have

$$\mathbb{E}[X^2] = np((n-1)p + 1)$$

and we have

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = np((n-1)p + 1) - n^2 p^2 = np(1-p)$$

Example 2.68. X is $P(\lambda)$ RV.

$$\begin{aligned} \mathbb{E}[X^2] &= \sum_{k=0}^{\infty} k^2 \frac{\lambda^k}{k!} e^{-\lambda} = \lambda \sum_{k=0}^{\infty} k \frac{\lambda^{k-1}}{(k-1)!} e^{-\lambda} \\ &= \lambda \sum_{l=0}^{\infty} (l+1) \frac{\lambda^l}{(l)!} e^{-\lambda} = \lambda(\lambda + 1) \end{aligned}$$

and

$$\text{Var}(X) = \lambda(\lambda + 1) - \lambda^2$$

EXERCISE: Compute the third, fourth moment for $B(p)$, $B(n, p)$, $P(\lambda)$.

Definition 2.69. Let X be a RV. The Moment Generating Function (MGF) of X is defined as

$$M(s) = \mathbb{E}[e^{sX}]$$

if $\mathbb{E}[e^{sX}]$ exists for s in some neighbourhood near 0, i.e. $\exists s_0 > 0$ such that $\mathbb{E}[e^{sX}] < \infty$ for every $s \in (-s_0, s_0)$.

$M(0) = 1$.

Remark 2.70. Is $X \in \{x_n : n \in \mathbb{N}\}$ then

$$M(s) = \mathbb{E}[e^{sX}] = \sum_n e^{sx_n} \mathbb{P}(X = x_n)$$

Proposition 2.71. If $M(s)$ exists for $s \in (-s_0, s_0)$ for some $s_0 > 0$, then $\mathbb{E}[X^k]$ exists $\forall k \in \{0, 1, 2, \dots\}$ and

$$M^{(k)}(0) = \frac{d^k}{ds^k} M(s)|_{s=0}$$

exists and

$$\mathbb{E}[X^k] = M^{(k)}(0)$$

Proof. $\forall k \in \{0, 1, 2, \dots\}; \forall s \in (-s_0, s_0), s \neq 0$ there exists $C_k > 0$ such that $|y_k| \leq C_k e^{|sy|}$ for all $y \in \mathbb{R}$. (polynomial growth can be controlled by exponential growth).

$$\sum_n |x_n|^k \mathbb{P}(X = x_n) \leq C_k \sum_n e^{|sy|} \mathbb{P}(X = x_n) \leq C_k \sum_n (e^{sx_n} + e^{-sx_n}) \mathbb{P}(X = x_n) = C_k (M(s) + M(-s))$$

Now

$$\frac{d^k}{ds^k} M(s) = \frac{d^k}{ds^k} \left(\sum_n e^{sx_n} \mathbb{P}(X = x_n) \right) \Big|_{s=0}$$

accept the next step

$$\begin{aligned} &= \sum_n \left(\frac{d^k}{ds^k} e^{sx_n} \right) \mathbb{P}(X = x_n) \Big|_{s=0} \\ &= \sum_n \left(x_n^k e^{sx_n} \right) \mathbb{P}(X = x_n) \Big|_{s=0} = \sum_n x_n^k \mathbb{P}(X = x_n) = \mathbb{E}[X^k] \end{aligned}$$

□

Remark 2.72. MGF does not always exist, e.g., $\mathbb{E}[|X|^k] = \infty$ for some k then MGF does not exist.

Even if $\mathbb{E}[X^k]$ exists $\forall k$, MGF does not have to exist.

Remark 2.73. If $\mathbb{E}[X^k]$ exists $\forall k$ and $\exists C > 0$ such that $|\mathbb{E}[X^k]| \leq k! C^k$ for all $k \in \mathbb{N}$ then MGF exists. In this case

$$M(s) = M(0) + M'(0)s + \frac{M''(0)}{2!} s^2 + \dots$$

$$M(s) = 1 + \mathbb{E}[X]s + \frac{\mathbb{E}[X^2]}{2!} s^2 + \dots$$

3 Variance, Moments, and Generating Functions

4 Multiple Random Variables: Correlation, Independence, Conditioning

5 The Law of Large Numbers